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MACHINE LEARNING • CASE STUDY

BANKRUPTCY PREDICTION

Using Decision Trees & Random Forest with Precision–Recall Analysis

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Presentation Roadmap

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Introduction

Domain

Corporate financial risk assessment uses financial indicators to understand a company's financial condition.

ML approach

Supervised classification learns patterns from labelled historical company records.

Why it matters

Early risk signals can help analysts prioritise review and investigate financial distress sooner.

- The system predicts the dataset label: bankrupt (1) or non-bankrupt (0).
- It compares an interpretable Decision Tree with an ensemble Random Forest.
- The project focuses on minority-class evaluation—not accuracy alone.

Problem Statement

CORE PROBLEM

How can we identify companies whose financial indicators resemble historical bankrupt companies—while controlling false alarms?

Existing challenge

Manual financial review can be slow and depends on analyst time and expertise.

Technical challenge

Financial indicators may have complex, non-linear relationships.

Class imbalance

Only 220 of 6,819 records are labelled bankrupt (≈3.23%).

Affected stakeholders: banks • lenders • investors • company analysts • risk teams

Motivation & Objectives

Motivation

Explore how ML can support early financial-risk screening and compare models under severe class imbalance.

Real-world relevance

A risk flag can prompt further financial review; it is not a guarantee or a final financial decision.

PROJECT OBJECTIVES

- Perform EDA and validate data quality.
- Use a stratified 70/30 train–test split.
- Select the top 10 features using training data only.
- Train and compare Decision Tree and Random Forest.
- Evaluate with precision, recall, F1, Average Precision and ROC-AUC.

Existing System & Limitations

Manual analysis

Analysts review balance sheets, ratios, cash flow and other financial statements.

Traditional scoring

Rule-based ratios or statistical scoring may simplify complex financial relationships.

Key limitations

Time-consuming; can vary by analyst; majority-class accuracy can hide missed bankrupt cases.

Project positioning

- This project is a reproducible ML case study, not a claim that all existing systems use only manual methods.
- It compares two tree-based classifiers and explicitly reports minority-class metrics.
- Model outputs should support—not replace—professional financial assessment.

Proposed System

A leakage-aware machine-learning pipeline that converts financial indicators into a predicted class and evaluation evidence.

Input

Company financial indicators from the Taiwanese Bankruptcy Prediction dataset.

Processing

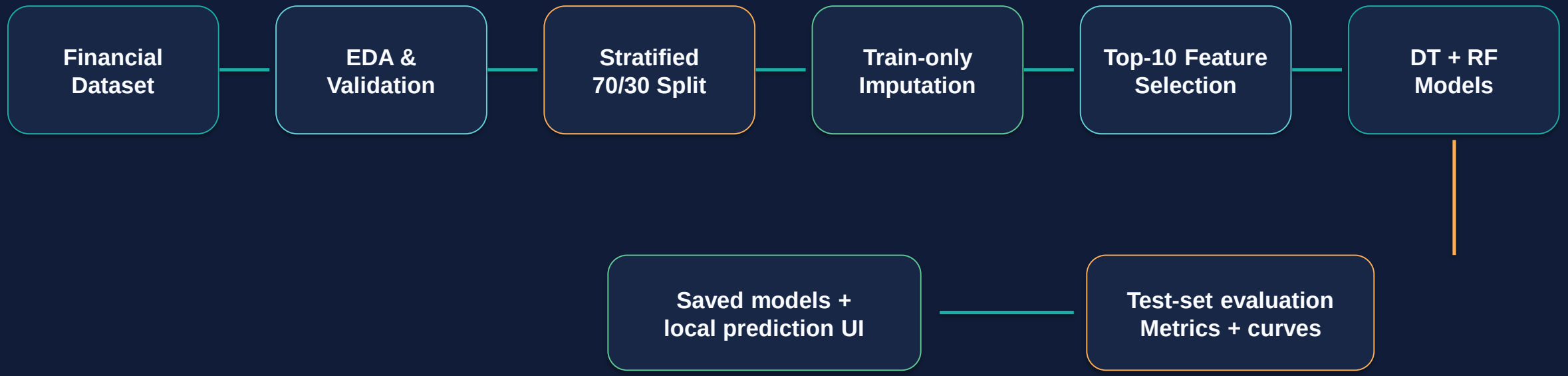
EDA → split → training-only imputation → feature selection → model training.

Output

Bankrupt / non-bankrupt prediction, probability output and evaluation charts.

Design principle: the held-out test set is not used to fit preprocessing or select features.

System Architecture / Block Diagram



Input: 95 financial features | Processing: preprocessing + feature selection | Output: class prediction and risk-analysis metrics

Methodology / Workflow

1. Load data
2. EDA & quality checks
3. Stratified split
4. Fit imputer on train
5. Select top 10 features
6. Train DT & RF
7. Evaluate test set
8. Save models & reports

Leakage prevention: test data is held out until final evaluation.

Technology Stack

Language

Python 3.x

Notebook

Google Colab / Jupyter

Data handling

Pandas • NumPy

ML

scikit-learn

Visualisation

Matplotlib

Model persistence

joblib / saved model artifacts

Interface

HTML • CSS • JavaScript • Python server

Dataset source

UCI Taiwanese Bankruptcy Prediction

System Modules / Components

Data loading

Read CSV/XLSX data and identify target.

EDA module

Summaries, class distribution, correlations.

Preprocessing

Training-only median imputation.

Feature selection

Rank 95 features; retain top 10.

ML module

Train Decision Tree and Random Forest.

Evaluation

Metrics, confusion matrices, PR curve.

Prediction UI

Use saved models for inference.

Reporting

Save charts, CSV metrics and EDA report.

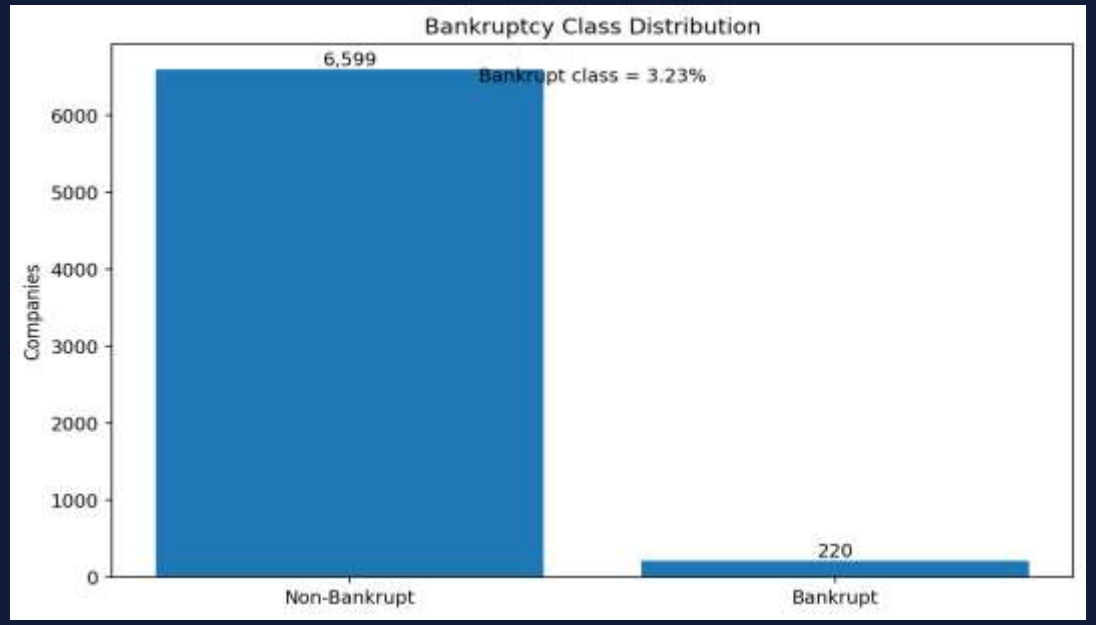
Dataset / Input Data

6,819
Company records

95
Financial input features

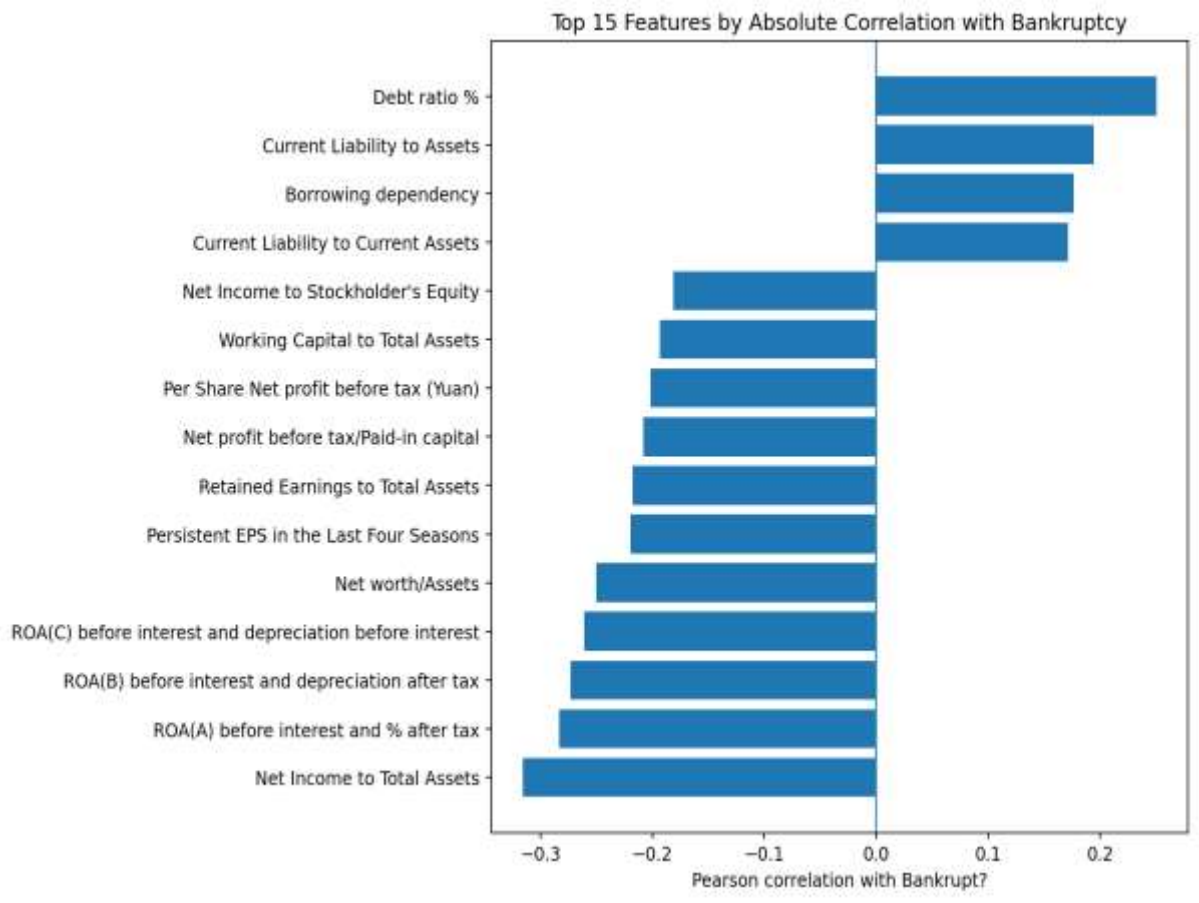
220
Bankrupt records

3.23%
Bankrupt class



- Target: Bankrupt? (0 = non-bankrupt, 1 = bankrupt)
- 6,599 non-bankrupt records; 220 bankrupt records.
- Missing values: 0 • Duplicate rows: 0
- Stratified split: 4,773 train / 2,046 test.

Exploratory Data Analysis (EDA)



What EDA checks

Dataset shape, missing values, duplicates and target distribution.

Correlation finding

Net Income to Total Assets had the strongest absolute correlation in the displayed top-15 analysis (about −0.3155).

Interpretation

Correlation indicates association, not causation; final features are selected on training data only.

Class imbalance is the key EDA finding and motivates precision–recall analysis.

Algorithm / Model

Decision Tree

- Single tree of threshold-based rules
- max_depth = 8
- class_weight = balanced
- Easier to explain; may be less stable

Random Forest

- Ensemble of many decision trees
- Final model: 300 trees, max_depth = 10
- class_weight = balanced
- n_jobs = −1 for parallel execution

Feature selector

A separate 500-tree Random Forest ranks all 95 features using training data only; the top 10 are retained.

random_state = 42 supports reproducible splits and model runs.

Algorithm / Model — Pseudocode

Decision Tree

1 TREE

- Threshold-based splits
- max_depth = 8 • class_weight = balanced

PSEUDOCODE

```
INPUT X_train, y_train
create root node
WHILE depth < 8:
    choose best feature + threshold
    split data into left / right nodes
assign class at leaf nodes
OUTPUT predicted class
```

Random Forest

300 TREES

- Bootstrap rows + random feature subsets
- max_depth = 10 • class_weight = balanced

PSEUDOCODE

```
FOR tree = 1 TO 300:
    draw bootstrap sample of train rows
    grow tree using random feature subsets
END FOR
collect predictions from all trees
majority vote / average probabilities
OUTPUT final class + probability
```

FEATURE SELECTION 500-tree Random Forest on TRAIN data only → rank 95 features → retain Top 10 → train final DT + RF

Top 10 Selected Features

01 Total debt / Total net worth

02 Borrowing dependency

03 Persistent EPS in the Last Four Seasons

04 Continuous interest rate (after tax)

05 Debt ratio %

06 Retained Earnings to Total Assets

07 Net worth / Assets

08 Equity to Liability

09 Liability to Equity

10 Net Income to Total Assets

Selection was performed after splitting the data, using training observations only.

Implementation Highlights

- Loaded and validated the target column and 95 numerical features.
- Generated EDA outputs: class distribution, correlations and dataset summary.
- Created a stratified train/test split using random_state = 42.
- Fitted median imputers on training data only.
- Selected the top 10 features using training-only Random Forest importance.
- Trained final Decision Tree and Random Forest models.
- Saved model artifacts, metrics, confusion matrices and Precision–Recall plots.
- Provided a local dashboard for predictions from saved models.

Important: the dashboard performs inference; it does not retrain the models for each prediction.

Results: Model Comparison

Metric	Decision Tree	Random Forest
Accuracy	91.94%	94.77%
Precision	22.35%	30.84%
Recall	60.61%	50.00%
F1-score	32.65%	38.15%
Average Precision	24.06%	30.57%
ROC-AUC	77.62%	91.23%

Best overall ranking

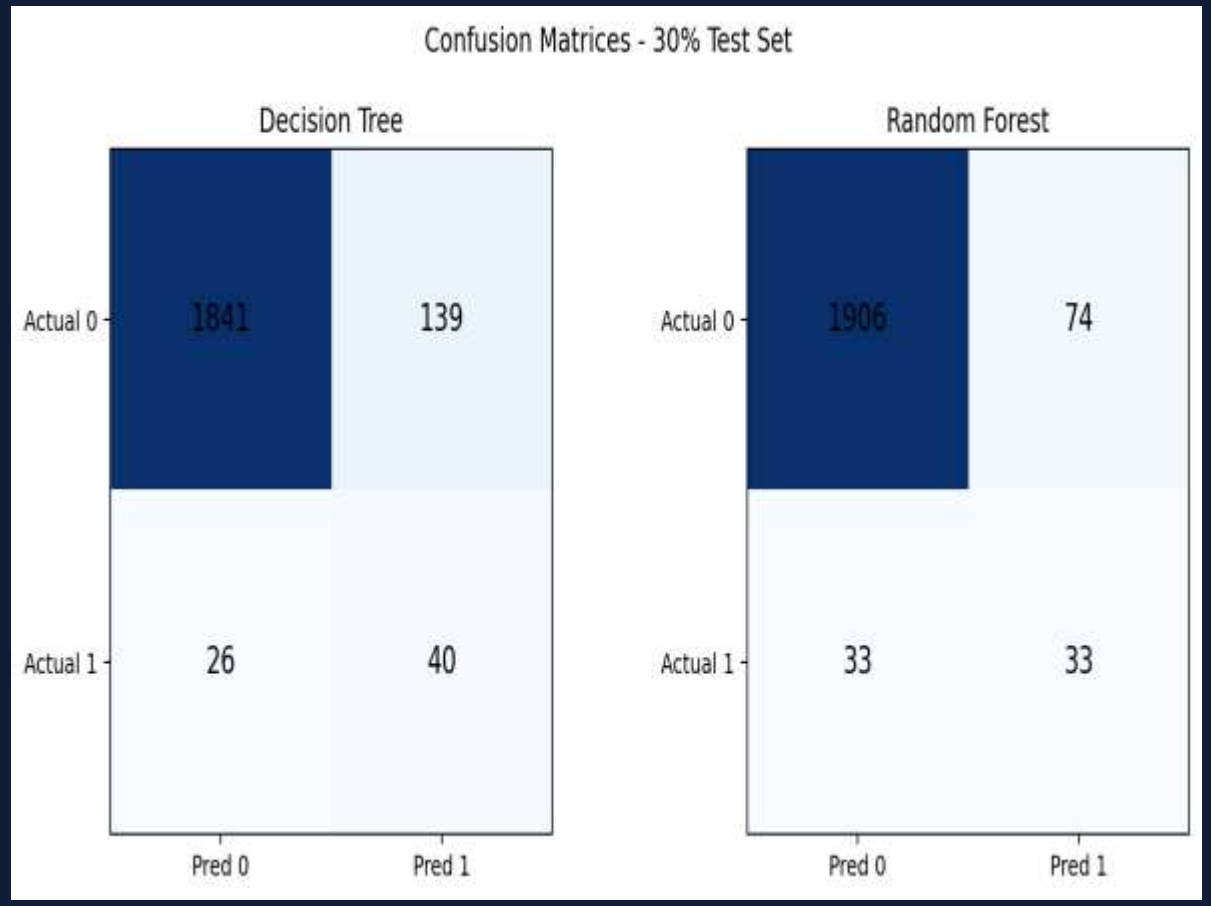
Random Forest: ROC-AUC 91.23%, Average Precision 30.57%.

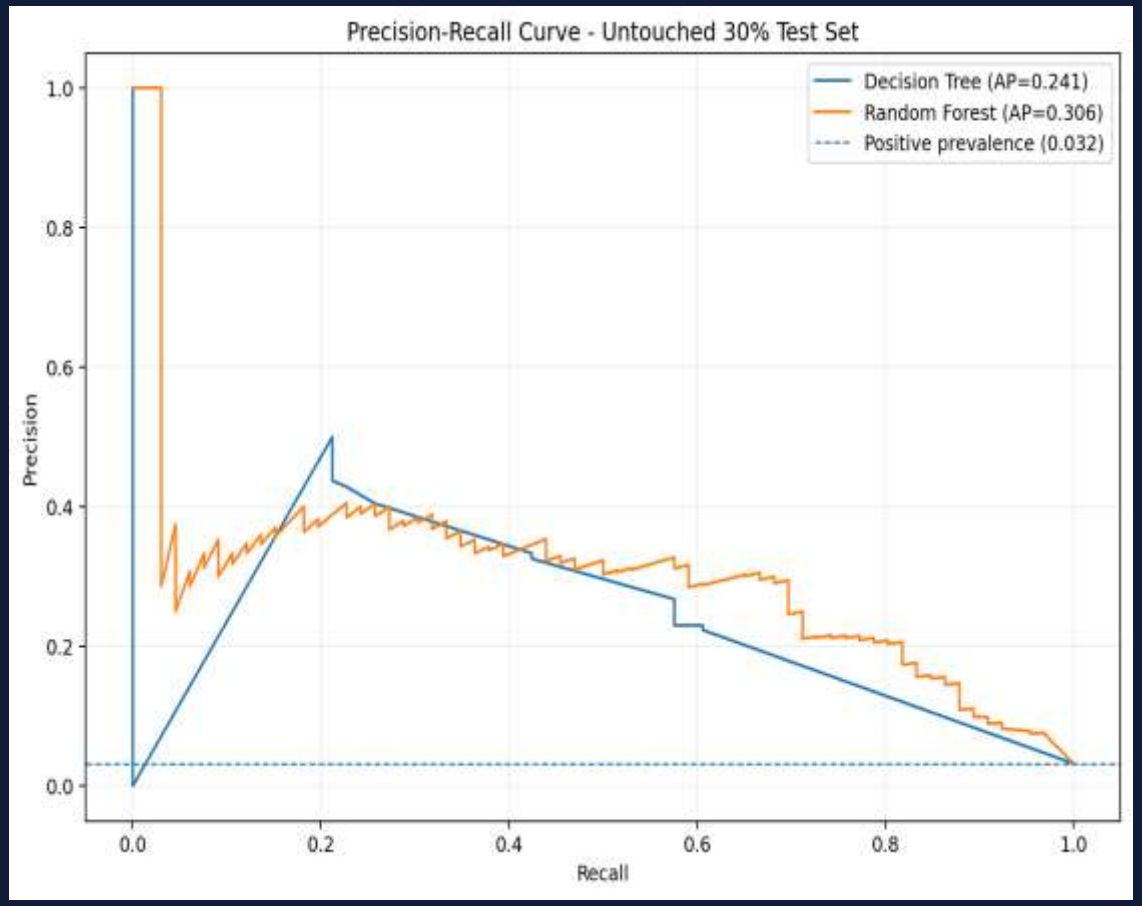
Higher recall

Decision Tree detects 60.61% of bankrupt test cases, compared with 50.00% for Random Forest.

No single model wins every metric; selection depends on the cost of missed cases vs false alarms.

Confusion Matrices & Precision–Recall



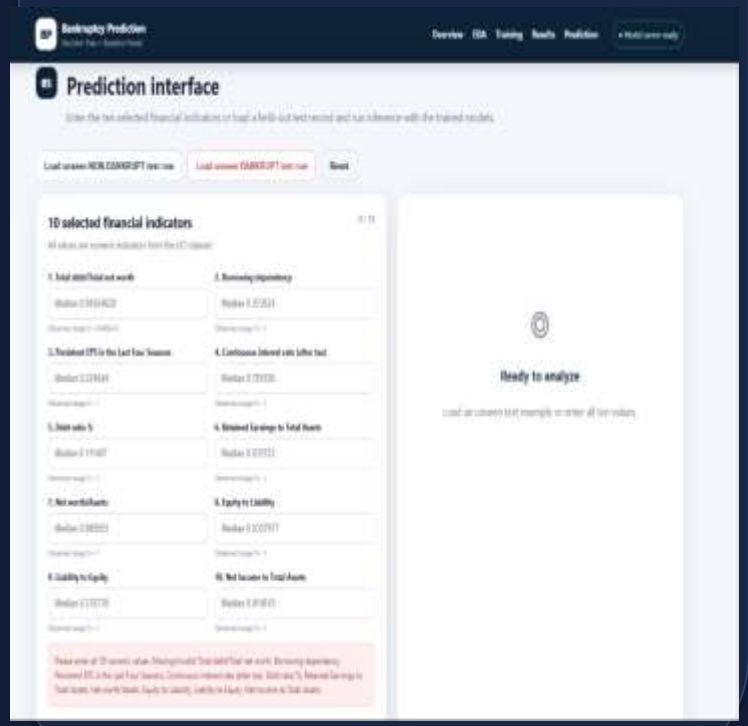


Decision Tree: TN 1841 • FP 139 • FN 26 • TP 40

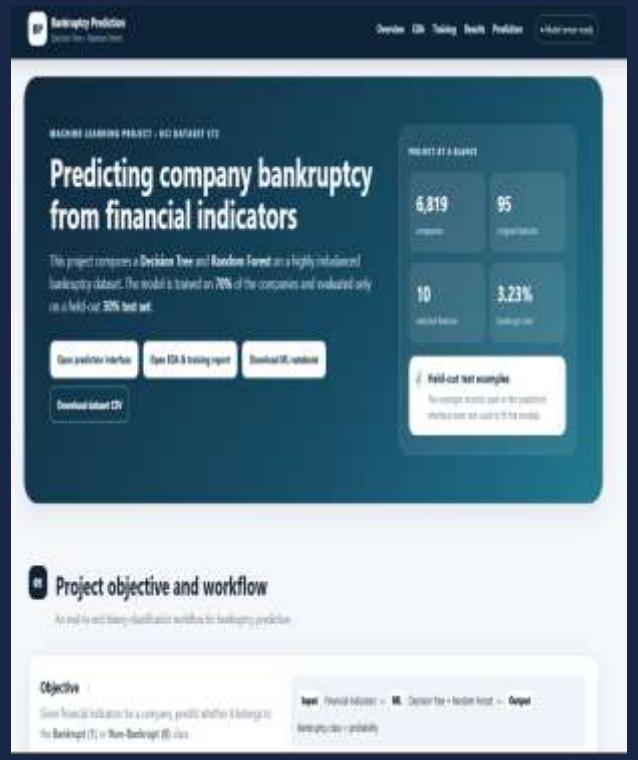
Random Forest: TN 1906 • FP 74 • FN 33 • TP 33

Screenshots / Demo

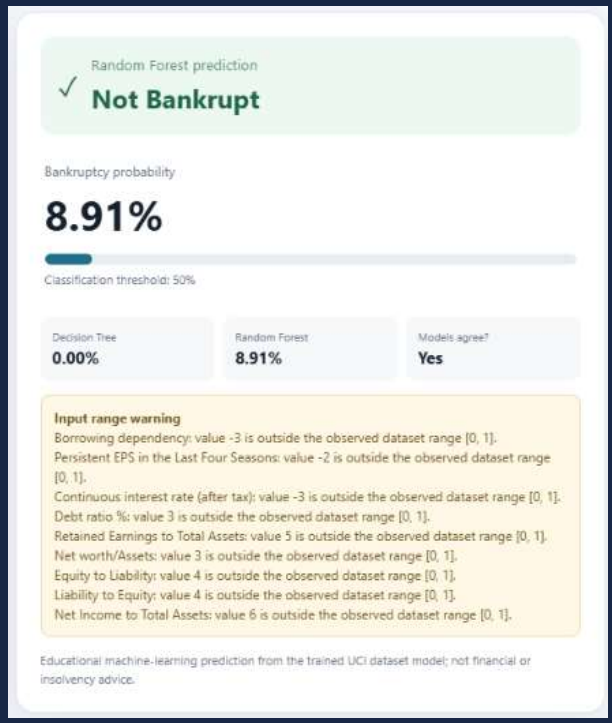
1 • Input screen



2 • Prediction



3 • Output



Replace this slide's panels with real application screenshots if your expo requires screen-by-screen evidence.

Advantages, Applications & Limitations

Advantages

- Reproducible ML workflow
- Compares two classifiers
- Training-only feature selection
- Multiple metrics for imbalance
- Saved models and report outputs

Applications

- Preliminary corporate risk screening
- Credit-risk analysis support
- Investment research support
- Financial analyst review queues
- Academic ML demonstrations

Limitations

- Historical dataset may not reflect current markets
- Only 220 bankrupt records
- Model performance may shift on new data
- No sole reliance for financial decisions

Responsible use: treat predictions as decision-support signals, with expert review and validation.

Future Scope

Data expansion

Validate with more recent and external company financial data.

Model improvement

Tune thresholds, compare more algorithms and calibrate probabilities.

Deployment

Add monitoring, explainability and secure, scalable deployment.

Any improvement should be tested on independent data before claiming better real-world performance.

Conclusion

What the project achieved

- Completed EDA on 6,819 records and 95 financial features.
- Built a leakage-aware train/test and feature-selection pipeline.
- Compared Decision Tree and Random Forest on unseen test data.
- Generated model metrics, confusion matrices and PR analysis.

Key result

Random Forest leads on most metrics; Decision Tree has higher bankrupt-class recall.

The project demonstrates ML-based financial-risk screening—not a guaranteed bankruptcy decision.

THANK YOU

Project Artifacts & References

- Dataset: UCI Taiwanese Bankruptcy Prediction (Dataset 572).
- Notebook: Bankruptcy_ML_Colab_Complete.ipynb.
- Training code: training/train_pipeline.py.
- EDA assets: class distribution, top correlations, feature importance, train/test split.
- Evaluation assets: model_metrics.csv, confusion_matrices.png, precision_recall_curve.png.
- Saved artifacts: selected Decision Tree, Random Forest and preprocessing objects.
- Report: Bankruptcy_EDA_Training_Report.pdf.

Note: performance values shown are from the project's saved evaluation artifacts.